Lecture 28

Power Series Solutions to Linear Differential Equations

In the previous lectures, we have begun to lay the groundwork in solving equations using power series. We have the computational ability to do so, but why are we guaranteed a power series solution to a differential equation with a given center?

Definition: Consider the homogeneous linear differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = 0. \tag{28.1}$$

In standard form it is written

$$y^{(n)} + P_1(x)y^{(n-1)} + \cdots + P_n(x)y = 0,$$

where $P_i(x) = \dfrac{a_{n-i}(x)}{a_n(x)}$ for $1 \leq i \leq n$. A point x_0 is called an **ordinary point** of (28.1) if $P_i(x)$ is analytic at x_0 for all $1 \leq i \leq n$.

Recall, has a power series representation at x_0.

If x_0 is not an ordinary point, it is called a **singular point** of the equation.

NOTE: We will mostly be consider 1st and 2nd-order equations when solving using power series, but the techniques and theory apply to higher order equations. The previous defintion in the 2nd-order case consists of the equation

$$y'' + P(x)y' + Q(x)y = 0$$

with x_0 ordinary if $P(x)$ and $Q(x)$ are analytic at x_0, and otherwise x_0 is a singular point.

Example: Let $x_0 = 0$. Is x_0 an ordinary or singular point of the following equations?
center

1. $\frac{1}{x^2}y'' - \frac{2}{x}y' + y = 0$ $\Rightarrow$ multiplying by x^2

$\Rightarrow y'' - 2xy' + x^2y = 0$

$P(x) = -2x$, $Q(x) = x^2$ polynomials so analytic at $x=0$.

So $x=0$ is an ordinary point.

2. $y'' + \cos(x)y' - e^x y = 0$

$\Rightarrow P(x) = \cos(x)$, $Q(x) = -e^x$ both analytic at $x=0$

so $x=0$ ordinary pt.

3. $x^2y'' + 4xy' + xy = 0$ $\Rightarrow y'' + \frac{4}{x}y' + \frac{1}{x}y = 0$

$P(x) = \frac{4}{x}$, $Q(x) = \frac{1}{x}$ DNE at $x=0$, not analytic

so $x=0$ is a singular pt.

4. $xy''' + (1-x)y'' - xy' + y = 0$

$\Rightarrow y''' + \frac{(1-x)}{x}y'' - y' + \frac{1}{x}y = 0$

$P_1 = \frac{1-x}{x}$, $P_2 = -1$, $P_3 = \frac{1}{x}$, P_1, P_3 not analytic at $x=0$

$x=0$ is a singular pt.

Example: Determine all singular points of

$$xy'' + \frac{x}{1-x}y' + \sin(x)y = 0$$

$$\Rightarrow \quad y'' + \frac{x}{x(1-x)}y' + \frac{\sin(x)}{x}y = 0$$

$$P(x) = \frac{\cancel{x}}{\cancel{x}(1-x)} = \frac{1}{1-x}, \quad x \neq 0$$

, hole at $x=0$ removable discontinuity

analytic at $x=0$.

We can assume that $P(x)$ is defined so it is analytic at $x=0$

no removable discontinuity.

$$Q(x) = \frac{\sin(x)}{x}, \quad \lim_{x\to 0^+} \frac{\sin(x)}{x} = 1, \quad \text{Recall: } \sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\text{So} \quad \frac{\sin(x)}{x} = \frac{\cancel{x} - \frac{x^{\cancel{3}\,2}}{3!} + \frac{x^{\cancel{5}\,4}}{5!} - \cdots}{\cancel{x}} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!}$$

So $\frac{\sin(x)}{x}$ is analytic at $x=0$.

$x=0$ is an ordinary pt.

$\boxed{x=1 \text{ is a singular pt.}}$

NOTE: To find a solution to an equation using a power series, we will need to prescribe a center for the series. As long as the center is an ordinary point, we can assume that a power series solution exists, but if the center is a singular point, there is no such guarantee. We will make this idea rigorous in the next lecture.

Method to Solve DEQ Using a Power Series

To solve $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$, assuming y is analytic at x_0, we use the following steps:

1. Let $y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$.

   a_n - coefficients
   x_0 - center

2. Use $y' = \sum_{n=1}^{\infty} na_n(x - x_0)^{n-1}$ and $y'' = \sum_{n=2}^{\infty} n(n-1)a_n(x - x_0)^{n-2}$ and substitute y, y', and y'' into the equation.
3. Find a recurrence relation for a_n.
4. Use any initial conditions if applicable.

Example: Consider the equation $y' - 2xy = 0$ at $x_0 = 0$.

$P(x) = -2x$, analytic at $x_0 = 0$. $x=0$ is an ordinary pt.

$$y' - 2xy = 0 \Rightarrow \underbrace{\sum_{n=1}^{\infty} na_n(x-0)^{n-1}}_{y'} - 2x\underbrace{\sum_{n=0}^{\infty} a_n(x-0)^n}_{y} = 0$$

$$\Rightarrow \sum_{n=1}^{\infty} na_n x^{n-1} - \sum_{n=0}^{\infty} 2a_n x^{n+1} = 0$$

up 1 down 1

$$\Rightarrow \sum_{n=0}^{\infty} (n+1)a_{n+1} x^n - \sum_{n=1}^{\infty} 2a_{n-1} x^n = 0$$

separate $n=0$ term

$$\Rightarrow (0+1)a_{0+1}x^0 + \sum_{n=1}^{\infty}(n+1)a_{n+1}x^n - \sum_{n=1}^{\infty} 2a_{n-1}x^n = 0$$

$$\Rightarrow \underset{n=0}{a_1} + \sum_{n=1}^{\infty}\left[(n+1)a_{n+1} - 2a_{n-1}\right]x^n = 0$$

Recall $\sum_{n=0}^{\infty} a_n(x-x_0)^n = 0$
$\Rightarrow a_n = 0$ for all n.

$$\Rightarrow \begin{cases} \boxed{a_1 = 0} & (n=0) \\ (n+1)a_{n+1} - 2a_{n-1} = 0, & n \geq 1 \end{cases}$$

recurrence relation.

Solve for a_{n+1} : $(n+1)a_{n+1} = 2a_{n-1} \Rightarrow \boxed{a_{n+1} = \frac{2}{n+1} a_{n-1},\ n \geq 1}$

Test:

$n=1$: $a_2 = \frac{2}{2} a_0 = a_0 = \frac{a_0}{1!}$ — constant.

$n=2$: $a_3 = \frac{2}{3} a_1 = \frac{2}{3} \cdot 0 = \boxed{0}$

$n=3$: $a_4 = \frac{2}{4} a_2 = \frac{2}{4} \cdot \frac{2}{2} a_0 = \frac{1}{2} a_0 = \frac{a_0}{2!}$

$n=4$: $a_5 = \frac{2}{5} a_3 = \frac{2}{5} \cdot 0 = \boxed{0}$

$n=5$: $a_6 = \frac{2}{6} a_4 = \frac{2}{6} \cdot \frac{1}{2} a_0 = \frac{1}{6} a_0 = \frac{a_0}{3!}$

$n=6$: $a_7 = \frac{2}{7} a_5 = \frac{2}{7} \cdot 0 = \boxed{0}$

$n=7$: $a_8 = \frac{2}{8} a_6 = \frac{2}{8} \cdot \frac{1}{6} a_0 = \frac{1}{24} a_0 = \frac{a_0}{4!}$

$\vdots$

$a_n = \begin{cases} \text{odd} & = 0 \\ \text{even} & \neq 0. \end{cases} \Rightarrow a_{2n} = \frac{a_0}{n!},\ n \geq 1$

So $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots$

$= a_0 + 0 \cdot x + \frac{a_0}{1!} x^2 + 0 x^3 + \frac{a_0}{2!} x^4 + \cdots$

$= a_0 \left[1 + \frac{1}{1!} x^2 + \frac{1}{2!} x^4 + \frac{1}{3!} x^6 + \cdots \right]$

$\Rightarrow \boxed{y = a_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}} = a_0 e^{x^2}$

$y' - 2xy = 0$

$\mu(x) = e^{\int -2x\,dx} = e^{-x^2}$, $e^{-x^2} y = \int 0 = C$

$\boxed{y = Ce^{x^2}}$

Recall, $y = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, $e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$

Example: Consider the equation $y'' + y = 0$ at $x_0 = 0$ $\longrightarrow$ $y_h = c_1\cos(x) + c_2\sin(x)$

use $y = \sum_{n=0}^{\infty} a_n x^n$, $y'' = \sum_{n=2}^{\infty} n(n-1)\, a_n x^{n-2}$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$$

up 2

down 2

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \Big[\underbrace{(n+2)(n+1) a_{n+2} + a_n}_{=0}\Big] x^n = 0$$

$$(n+2)(n+1) a_{n+2} + a_n = 0 \Rightarrow (n+2)(n+1) a_{n+2} = -a_n$$

$$\Rightarrow \boxed{a_{n+2} = \frac{-1}{(n+2)(n+1)} a_n, \quad n \geq 0}$$

recurrence relation.

A shift of 2, $(n \to n+2)$ splits evens and odds.

<u>Evens</u>:

$n=0$: $a_2 = \frac{-1}{2\cdot 1} a_0 = \frac{-1}{2!} a_0$

$n=2$: $a_4 = \frac{-1}{4\cdot 3} a_2 = \frac{-1}{4\cdot 3} \cdot \frac{-1}{2!} a_0$

$= \frac{(-1)^2}{4\cdot 3\cdot 2\cdot 1} a_0$

$= \frac{(-1)^2}{4!} a_0$

$n=4$: $a_6 = \frac{-1}{6\cdot 5} a_4 = \frac{-1}{6\cdot 5} \cdot \frac{(-1)^2}{4!} a_0$

<u>Odds</u>:

$n=1$: $a_3 = \frac{-1}{3\cdot 2} a_1 = \frac{-1}{3!} a_1$

$n=3$: $a_5 = \frac{-1}{5\cdot 4} a_3 = \frac{-1}{5\cdot 4} \cdot \frac{-1}{3!} a_1$

$= \frac{(-1)^2}{5!} a_1$

$n=5$: $a_7 = \frac{-1}{7\cdot 6} a_5$

$= \frac{-1}{7\cdot 6} \cdot \frac{(-1)^2}{5!} a_1$

$$= \frac{(-1)^3}{6!} a_0 \qquad = \frac{(-1)^3}{7!} a_1$$

If $n = 2k$

$$a_2 = a_{2\cdot 1} = \frac{-1}{2b} a_0 = \frac{(-1)^k}{2\cdot k!} a_0$$

$k=1$

$$a_n = a_{2k} = \frac{(-1)^k}{(2k)!} a_0$$

If $n = 2k+1$

$$a_n = a_{2k+1} = \frac{(-1)^k}{(2k+1)!} a_1$$

$$y = \sum_{n=0}^{\infty} a_n x^n = \underbrace{\sum_{n=0}^{\infty} a_{2n} x^{2n}}_{\text{evens}} + \underbrace{\sum_{n=0}^{\infty} a_{2n+1} x^{2n+1}}_{\text{odds}}$$

$$= \sum_{n=0}^{\infty} a_0 \frac{(-1)^n}{(2n)!} x^{2n} + \sum_{n=0}^{\infty} a_1 \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\Rightarrow \boxed{y = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + a_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}}$$

$$= a_0 \cos(x) + a_1 \sin(x)$$

Example: Consider the equation $2y'' + xy' + y = 0$ at $x_0 = 0$

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substitute

$$2\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + x\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0$$

up 2

$$\Rightarrow \sum_{n=2}^{\infty} 2n(n-1)a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

down 2

$$\Rightarrow \sum_{n=0}^{\infty} 2(n+2)(n+1)a_{n+2} x^n + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

separate $n=0$ terms

$$\Rightarrow \left[2(0+2)(0+1)a_{0+2}x^0 + a_0 x^0\right] + \sum_{n=1}^{\infty}\left[2(n+2)(n+1)a_{n+2} + n a_n + a_n\right]x^n = 0$$

$$\Rightarrow \underbrace{\left[4a_2 + a_0\right]}_{=0} + \sum_{n=1}^{\infty}\underbrace{\left[2(n+2)(n+1)a_{n+2} + (n+1)a_n\right]}_{=0} x^n = 0$$

$$\Rightarrow \begin{cases} 4a_2 + a_0 = 0 & (n=0) \\ 2(n+2)(n+1)a_{n+2} + (n+1)a_n = 0, & n \geq 1 \end{cases}$$

$$\Rightarrow 4a_2 + a_0 = 0 \Rightarrow 4a_2 = -a_0 \Rightarrow a_2 = -\frac{1}{4}a_0 \quad (n=0)$$

$$2(n+2)(n+1)a_{n+2} + (n+1)a_n = 0$$

$$\Rightarrow 2(n+2)\cancel{(n+1)}a_{n+2} = -\cancel{(n+1)}a_n \quad \text{recurrence relation}$$

$$\Rightarrow \boxed{a_{n+2} = \frac{-1}{2(n+2)}a_n, \quad n \geq 1}$$

shift of 2

<u>Evens</u>:

$n=0$: $\quad a_2 = -\frac{1}{4\cdot 1}a_0 = \frac{-1}{1!\,2^2}a_0$

$n=2$: $\quad a_4 = \frac{-1}{2(4)}a_2 = \frac{-1}{2\cdot 4}\cdot\frac{-1}{4}a_0 = \frac{(-1)^2}{2\cdot 4\cdot 4}a_0$

$$= \frac{(-1)^2}{2!\,2^4}a_0$$

$n=4$: $\quad a_6 = \frac{-1}{2(6)}a_4 = \frac{-1}{2\cdot 6}\cdot\frac{(-1)^2}{2\cdot 4\cdot 4}a_0 = \frac{(-1)^3}{2\cdot 6\cdot 2\cdot 4\cdot 4}a_0$

$$= \frac{(-1)^3}{3!\;2^6}a_0$$

$n=6$: $\quad a_8 = \frac{-1}{2\cdot 8}a_6 = \frac{-1}{2\cdot 8}\,\frac{(-1)^3}{3!\,2^6}a_0 = \frac{(-1)^4}{4!\,2^8}a_0$

$2\cdot 4$

$$\boxed{a_{2k} = \frac{(-1)^k}{k!\;2^{2k}}a_0}$$

<u>Odds</u>:

$$n=1: \quad a_3 = \frac{-1}{2\cdot 3} a_1 = \frac{(-1)^1}{2\,(3)} a_1$$

$$n=3: \quad a_5 = \frac{-1}{2\cdot 5} a_3 = \frac{-1}{2\cdot 5}\cdot\frac{-1}{2\cdot 3} a_1 = \frac{(-1)^2}{2^2(3\cdot 5)} a_1$$

$$n=5: \quad a_7 = \frac{-1}{2\cdot 7} a_5 = \frac{-1}{2\cdot 7}\cdot\frac{-1}{2\cdot 5}\cdot\frac{-1}{2\cdot 3} a_1$$

$$= \frac{(-1)^3}{2^3(3\cdot 5\cdot 7)} a_1$$

$$n=7: \quad a_9 = \frac{-1}{2\cdot 9} a_7 = \frac{-1}{2\cdot 9}\cdot\frac{(-1)^3}{2^3(3\cdot 5\cdot 7)} a_1$$

$$= \frac{(-1)^4}{2^4(3\cdot 5\cdot 7\cdot 9)} a_1$$

$$\vdots$$

$$a_{2k+1} = \frac{(-1)^k}{2^k\underbrace{(3\cdot 5\cdot 7\cdot\cdots\cdot(2k+1))}_{(2k+1)!!}} a_1$$

<u>Notation</u>:

$n! = n(n-1)! = n(n-1)(n-2)\cdots 2\cdot 1$, $\quad n!! = \begin{cases} n(n-2)(n-4)\cdots(2) & \text{even} \\ n(n-2)(n-4)\cdots(1) & \text{odd} \end{cases}$

1 $\quad n=0$

So $$\boxed{y = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} a_0 \frac{(-1)^n}{n!\,2^{2n}} x^{2n} + \sum_{n=0}^{\infty} a_1 \frac{(-1)^n}{2^n((2n+1)!!)} x^{2n+1}}$$

<u>Recall</u>: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$$a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,2^{2n}} x^{2n} = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^n}{n!\,(2^2)^n}$$

$$= a_0 \sum_{n=0}^{\infty} \frac{\left(\frac{-x^2}{4}\right)^n}{n!} = a_0 e^{-\frac{x^2}{4}}$$

So $y = e^{-x^2/4}$ solves the equation.

$$y = a_0 e^{-x^2/4} + a_1 \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{2^n((2n+1)!!)} x^{2n+1}}_{\text{function obtained by reduction of order using } y_1 = e^{-x^2/4}}$$

form $\sim\sim e^{-x^2/4} \operatorname{erf}(\sim\sim)$

error function

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,dt$$

Example: For the IVP $y'' + (x-2)y' - y = 0$, $y(0) = -1$, $y'(0) = 0$, find the first 4 nonzero terms in a power series solution centered at $x_0 = 0$.

If $y = \sum_{n=0}^{\infty} a_n x^n$, $y(0) = -1$

$= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots$, $y(0) = \boxed{a_0 = -1}$

$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = a_1 + 2a_2 x + 3a_3 x^2 + \cdots$

$y'(0) = \boxed{a_1 = 0}$

$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$

Substitute:

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + (x-2)\sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=1}^{\infty} n a_n x^n - \sum_{n=1}^{\infty} 2n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

up 2 down 2 up 1 1 down

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} 2(n+1) a_{n+1} x^n - \sum_{n=0}^{\infty} a_n x^n = 0$$

separate $n=0$ terms

$$\underbrace{[2a_2 - 2a_1 - a_0]}_{=0} x^0 + \sum_{n=1}^{\infty} \underbrace{[(n+2)(n+1)a_{n+2} + n a_n - 2(n+1)a_{n+1} - a_n]}_{=0} x^n = 0$$

$$\Rightarrow \begin{cases} 2a_2 - 2a_1 - a_0 = 0 & (n=0) \\ (n+2)(n+1)a_{n+2} - 2(n+1)a_{n+1} + (n-1)a_n = 0, & n \geq 1 \end{cases}$$

$$2a_2 - 2a_1 - a_0 = 0 \Rightarrow 2a_2 = 2a_1 + a_0$$
$$\Rightarrow 2a_2 = 2\cdot 0 + (-1)$$
$$\Rightarrow 2a_2 = -1 \Rightarrow \boxed{a_2 = -\frac{1}{2}}$$

$$(n+2)(n+1)a_{n+2} - 2(n+1)a_{n+1} + (n-1)a_n = 0$$
$$\Rightarrow (n+2)(n+1)a_{n+2} = 2(n+1)a_{n+1} - (n-1)a_n$$
$$\Rightarrow \boxed{a_{n+2} = \frac{2(n+1)a_{n+1} - (n-1)a_n}{(n+2)(n+1)}, \ n \geq 1}$$

recurrence relation.

$$n=1: \quad a_3 = \frac{2(2)a_2 - 0\,a_1}{3\cdot 2} = \frac{4}{6}a_2 = \frac{4}{6}\cdot\frac{-1}{2} = \frac{-1}{3}$$
$$\Rightarrow \boxed{a_3 = \frac{-1}{3}}$$

$$n=2: \quad a_4 = \frac{2(3)a_3 - 1\,a_2}{4\cdot 3} = \frac{6\cdot(-\frac{1}{3}) - (-\frac{1}{2})}{12}$$
$$= \frac{-2 + \frac{1}{2}}{12} = \frac{-\frac{3}{2}}{12} = -\frac{1}{8}$$
$$\Rightarrow \boxed{a_4 = -\frac{1}{8}}$$

So $y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$

$$= \boxed{-1 + 0\cdot x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{8}x^4} + \cdots$$